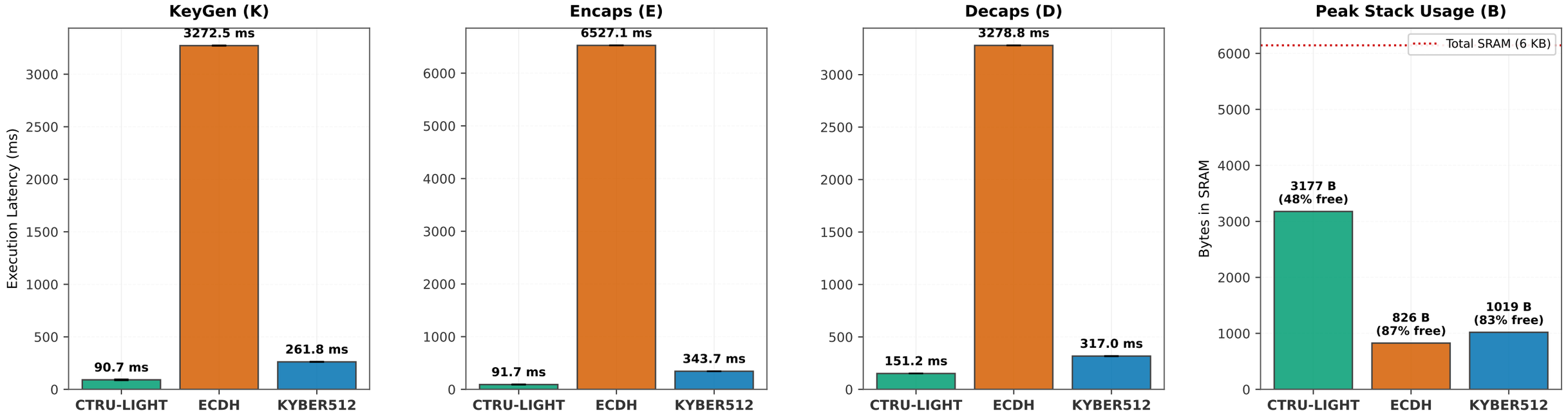
Cross-Algorithm Performance & Memory Benchmark on Physical ATmega4809 (20 MHz, ASM, -O3)
Direct hardware comparison between Post-Quantum KEMs (Kyber-512, CTRU-Light) and Classical ECDH (secp256r1 / micro-ecc).
KeyGen (K)
Execution Latency (ms)
3272.5 ms
90.7 ms
261.8 ms
CTRU-LIGHT
ECDH
KYBER512
Encaps (E)
6527.1 ms
91.7 ms
343.7 ms
Decaps (D)
3278.8 ms
151.2 ms
317.0 ms
Peak Stack Usage (B)
Bytes in SRAM
Total SRAM (6 KB)
3177 B
(48% free)
826 B
(87% free)
1019 B
(83% free)